

* Probability

* Poisson Distribution

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\lambda = np$$

* Normal distribution

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \bar{x}}{\sigma} \right)^2}, \quad -\infty < x < \infty$$

$$Z = \frac{x - \bar{x}}{\sigma} \quad \text{--- (S.N.V.)}$$

$\bar{x}$ = mean

σ = std deviation

* Moment Generating Function (MGF)

$$M_x(t) = \sum e^{tx} p(x)$$

* Continuous Random variable

$$H_r = \int_{-\infty}^{\infty} x^r f(x) dx$$

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Statistical Techniques

* Correlation Coefficient

$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\bar{y} = \frac{\sum y_i}{n}$$

$$r_{xy} = \frac{n \sum u_i v_i - (\sum u_i)(\sum v_i)}{\sqrt{[n \sum u_i^2 - (\sum u_i)^2][n \sum v_i^2 - (\sum v_i)^2]}}$$

$$u_i = \frac{x_i - a}{n}$$

$$v_i = \frac{y_i - b}{n}$$

* Rank correlation coefficient

$$R = 1 - \frac{6 \sum D^2}{N^3 - N}$$

For repeated Rank

$$R = 1 - \frac{6 \left[\sum D^2 + \frac{1}{12} (m_1^3 - m_1) + \frac{1}{12} (m_2^3 - m_2) \right]}{N^3 - N}$$

* Line of regression of x on y is

$$x - \bar{x} = b_{xy} (y - \bar{y})$$

$$b_{xy} = \frac{n \sum u_i v_i - (\sum u_i)(\sum v_i)}{n \sum v_i^2 - (\sum v_i)^2}$$

Fourier Series

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1) Interval $(0, 2\pi)$ and $(-\pi, \pi)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

2) Interval $(0, 2l)$ & $(-l, l)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$$

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

3) Fourier series for even & odd Function.
Interval $(-\pi, \pi)$

even

$$f(x) = 2 \int_0^{\pi} f(x) dx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

$$b_n = 0$$

odd

$$f(x) = 0$$

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A+B) - \sin(A-B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$$

$$\sin A \sin B = -\frac{1}{2} [\cos(A+B) - \cos(A-B)]$$

$$\int e^{ax} \sin bx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$\int e^{ax} \cos bx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

* Line of Regression of y on x

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$b_{yx} = \frac{n \sum u_i v_i - (\sum u_i)(\sum v_i)}{n \sum u_i^2 - (\sum u_i)^2}$$

* Equation of straight line

$$y_i = a u_i + b \quad \text{--- (1)}$$

$$u_i = \frac{x_i - a}{h}$$

$$a = \bar{x} = \frac{\sum x_i}{n}$$

taking $\sum$ on both side in eqⁿ (1)

$$\sum y_i = a \sum u_i + nb \quad \text{--- (2)}$$

multiply u_i and $\sum$ on both side

$$\sum u_i v_i = a \sum u_i^2 + b \sum u_i \quad \text{--- (3)}$$